

Newman-Penrose Formalism Calculations

Stephani et al., Equation (17.5)

Metric

$$ds^2 = -\epsilon ndx \otimes dx + m dx \otimes dy + m dy \otimes dx + l dy \otimes dy + e^M dz \otimes dz + \epsilon e^M dt \otimes dt$$

Coordinates

$$(x^\mu) = (x, y, z, t)$$

Metric Functions

$$n(t, z) = n$$

$$m(t, z) = m$$

$$l(t, z) = l$$

$$M(t, z) = M$$

Null Tetrad

Covariant components

$$l = \left(\frac{1}{2} \sqrt{2} \sqrt{\epsilon} \sqrt{n} \right) dx + \left(\frac{1}{2} \sqrt{2} \sqrt{\frac{m^2}{\epsilon n} + l} - \frac{\sqrt{2} m}{2 \sqrt{\epsilon} \sqrt{n}} \right) dy$$

$$n = \left(\frac{1}{2} \sqrt{2} \sqrt{\epsilon} \sqrt{n} \right) dx + \left(-\frac{1}{2} \sqrt{2} \sqrt{\frac{m^2}{\epsilon n} + l} - \frac{\sqrt{2} m}{2 \sqrt{\epsilon} \sqrt{n}} \right) dy$$

$$m = \left(\frac{1}{2} \sqrt{2} e^{\left(\frac{1}{2} M\right)} \right) dz + \left(\frac{1}{2} i \sqrt{2} \sqrt{\epsilon} e^{\left(\frac{1}{2} M\right)} \right) dt$$

$$\bar{m} = \left(\frac{1}{2} \sqrt{2} e^{\left(\frac{1}{2} M\right)} \right) dz + \left(-\frac{1}{2} i \sqrt{2} \sqrt{\epsilon} e^{\left(\frac{1}{2} M\right)} \right) dt$$

Contravariant components

$$l = \left(-\frac{\sqrt{\epsilon} l \sqrt{n}}{\sqrt{2\epsilon} l n + \sqrt{2} m^2} - \frac{m^2}{(\sqrt{2\epsilon} l n + \sqrt{2} m^2) \sqrt{\epsilon} \sqrt{n}} + \frac{\sqrt{\epsilon} l n + m^2 m}{(\sqrt{2\epsilon} l n + \sqrt{2} m^2) \sqrt{\epsilon} \sqrt{n}} \right) dx + \left(\frac{\sqrt{\epsilon} l n + m^2 \sqrt{\epsilon} \sqrt{n}}{\sqrt{2\epsilon} l n + \sqrt{2} m^2} \right) dy$$

$$n = \left(-\frac{\sqrt{\epsilon} l \sqrt{n}}{\sqrt{2\epsilon} l n + \sqrt{2} m^2} - \frac{m^2}{(\sqrt{2\epsilon} l n + \sqrt{2} m^2) \sqrt{\epsilon} \sqrt{n}} - \frac{\sqrt{\epsilon} l n + m^2 m}{(\sqrt{2\epsilon} l n + \sqrt{2} m^2) \sqrt{\epsilon} \sqrt{n}} \right) dx + \left(-\frac{\sqrt{\epsilon} l n + m^2 \sqrt{\epsilon} \sqrt{n}}{\sqrt{2\epsilon} l n + \sqrt{2} m^2} \right) dy$$

$$m = \left(\frac{1}{2} \sqrt{2} e^{\left(-\frac{1}{2} M\right)} \right) dz + \left(\frac{i \sqrt{2} e^{\left(-\frac{1}{2} M\right)}}{2 \sqrt{\epsilon}} \right) dt$$

$$\bar{m} = \left(\frac{1}{2} \sqrt{2} e^{\left(-\frac{1}{2} M\right)} \right) dz + \left(-\frac{i \sqrt{2} e^{\left(-\frac{1}{2} M\right)}}{2 \sqrt{\epsilon}} \right) dt$$

Directional Derivatives

$$D = -\frac{\sqrt{\epsilon} l \sqrt{n}}{\sqrt{2\epsilon} l n + \sqrt{2} m^2} - \frac{m^2}{(\sqrt{2\epsilon} l n + \sqrt{2} m^2) \sqrt{\epsilon} \sqrt{n}} + \frac{\sqrt{\epsilon} l n + m^2 m}{(\sqrt{2\epsilon} l n + \sqrt{2} m^2) \sqrt{\epsilon} \sqrt{n}} \partial_x + \frac{\sqrt{\epsilon} l n + m^2 \sqrt{\epsilon} \sqrt{n}}{\sqrt{2\epsilon} l n + \sqrt{2} m^2} \partial_y$$

$$\Delta = -\frac{\sqrt{\epsilon} l \sqrt{n}}{\sqrt{2\epsilon} l n + \sqrt{2} m^2} - \frac{m^2}{(\sqrt{2\epsilon} l n + \sqrt{2} m^2) \sqrt{\epsilon} \sqrt{n}} - \frac{\sqrt{\epsilon} l n + m^2 m}{(\sqrt{2\epsilon} l n + \sqrt{2} m^2) \sqrt{\epsilon} \sqrt{n}} \partial_x - \frac{\sqrt{\epsilon} l n + m^2 \sqrt{\epsilon} \sqrt{n}}{\sqrt{2\epsilon} l n + \sqrt{2} m^2} \partial_y$$

$$\delta = \frac{1}{2} \sqrt{2} e^{\left(-\frac{1}{2} M\right)} \partial_z + \frac{i \sqrt{2} e^{\left(-\frac{1}{2} M\right)}}{2 \sqrt{\epsilon}} \partial_t$$

$$\bar{\delta} = \frac{1}{2} \sqrt{2} e^{(-\frac{1}{2} M)} \partial_z - \frac{i \sqrt{2} e^{(-\frac{1}{2} M)}}{2 \sqrt{\epsilon}} \partial_t$$

Spin Coefficients

$$\rho = 0$$

$$\sigma = 0$$

$$\begin{aligned} \kappa = & \frac{\sqrt{2} \epsilon n^2 \frac{\partial}{\partial z} l}{8 \left(\epsilon e^{(\frac{1}{2} M)} l n^2 + e^{(\frac{1}{2} M)} m^2 n \right)} - \frac{\sqrt{2} \epsilon l n \frac{\partial}{\partial z} n}{8 \left(\epsilon e^{(\frac{1}{2} M)} l n^2 + e^{(\frac{1}{2} M)} m^2 n \right)} + \frac{i \sqrt{2} \sqrt{\epsilon} n^2 \frac{\partial}{\partial t} l}{8 \left(\epsilon e^{(\frac{1}{2} M)} l n^2 + e^{(\frac{1}{2} M)} m^2 n \right)} - \frac{i \sqrt{2} \sqrt{\epsilon} l n \frac{\partial}{\partial t} n}{8 \left(\epsilon e^{(\frac{1}{2} M)} l n^2 + e^{(\frac{1}{2} M)} m^2 n \right)} \\ & + \frac{\sqrt{2} m n \frac{\partial}{\partial z} m}{4 \left(\epsilon e^{(\frac{1}{2} M)} l n^2 + e^{(\frac{1}{2} M)} m^2 n \right)} - \frac{\sqrt{2} m^2 \frac{\partial}{\partial z} n}{4 \left(\epsilon e^{(\frac{1}{2} M)} l n^2 + e^{(\frac{1}{2} M)} m^2 n \right)} + \frac{i \sqrt{2} m n \frac{\partial}{\partial t} m}{4 \left(\epsilon e^{(\frac{1}{2} M)} l n^2 + e^{(\frac{1}{2} M)} m^2 n \right) \sqrt{\epsilon}} - \frac{\sqrt{2} \sqrt{\epsilon} l n + m^2 n \frac{\partial}{\partial z} m}{4 \left(\epsilon e^{(\frac{1}{2} M)} l n^2 + e^{(\frac{1}{2} M)} m^2 n \right)} \\ & - \frac{i \sqrt{2} m^2 \frac{\partial}{\partial t} n}{4 \left(\epsilon e^{(\frac{1}{2} M)} l n^2 + e^{(\frac{1}{2} M)} m^2 n \right) \sqrt{\epsilon}} + \frac{\sqrt{2} \sqrt{\epsilon} l n + m^2 m \frac{\partial}{\partial z} n}{4 \left(\epsilon e^{(\frac{1}{2} M)} l n^2 + e^{(\frac{1}{2} M)} m^2 n \right)} - \frac{i \sqrt{2} \sqrt{\epsilon} l n + m^2 n \frac{\partial}{\partial t} m}{4 \left(\epsilon e^{(\frac{1}{2} M)} l n^2 + e^{(\frac{1}{2} M)} m^2 n \right) \sqrt{\epsilon}} + \frac{i \sqrt{2} \sqrt{\epsilon} l n + m^2 m \frac{\partial}{\partial t} n}{4 \left(\epsilon e^{(\frac{1}{2} M)} l n^2 + e^{(\frac{1}{2} M)} m^2 n \right) \sqrt{\epsilon}} \end{aligned}$$

$$\mu = 0$$

$$\lambda = 0$$

$$\begin{aligned} \nu = & -\frac{\sqrt{2} \epsilon n^2 \frac{\partial}{\partial z} l}{8 \left(\epsilon e^{(\frac{1}{2} M)} l n^2 + e^{(\frac{1}{2} M)} m^2 n \right)} + \frac{\sqrt{2} \epsilon l n \frac{\partial}{\partial z} n}{8 \left(\epsilon e^{(\frac{1}{2} M)} l n^2 + e^{(\frac{1}{2} M)} m^2 n \right)} + \frac{i \sqrt{2} \sqrt{\epsilon} n^2 \frac{\partial}{\partial t} l}{8 \left(\epsilon e^{(\frac{1}{2} M)} l n^2 + e^{(\frac{1}{2} M)} m^2 n \right)} - \frac{i \sqrt{2} \sqrt{\epsilon} l n \frac{\partial}{\partial t} n}{8 \left(\epsilon e^{(\frac{1}{2} M)} l n^2 + e^{(\frac{1}{2} M)} m^2 n \right)} \\ & - \frac{\sqrt{2} m n \frac{\partial}{\partial z} m}{4 \left(\epsilon e^{(\frac{1}{2} M)} l n^2 + e^{(\frac{1}{2} M)} m^2 n \right)} + \frac{\sqrt{2} m^2 \frac{\partial}{\partial z} n}{4 \left(\epsilon e^{(\frac{1}{2} M)} l n^2 + e^{(\frac{1}{2} M)} m^2 n \right)} + \frac{i \sqrt{2} m n \frac{\partial}{\partial t} m}{4 \left(\epsilon e^{(\frac{1}{2} M)} l n^2 + e^{(\frac{1}{2} M)} m^2 n \right) \sqrt{\epsilon}} - \frac{\sqrt{2} \sqrt{\epsilon} l n + m^2 n \frac{\partial}{\partial z} m}{4 \left(\epsilon e^{(\frac{1}{2} M)} l n^2 + e^{(\frac{1}{2} M)} m^2 n \right)} \\ & - \frac{i \sqrt{2} m^2 \frac{\partial}{\partial t} n}{4 \left(\epsilon e^{(\frac{1}{2} M)} l n^2 + e^{(\frac{1}{2} M)} m^2 n \right) \sqrt{\epsilon}} + \frac{\sqrt{2} \sqrt{\epsilon} l n + m^2 m \frac{\partial}{\partial z} n}{4 \left(\epsilon e^{(\frac{1}{2} M)} l n^2 + e^{(\frac{1}{2} M)} m^2 n \right)} + \frac{i \sqrt{2} \sqrt{\epsilon} l n + m^2 n \frac{\partial}{\partial t} m}{4 \left(\epsilon e^{(\frac{1}{2} M)} l n^2 + e^{(\frac{1}{2} M)} m^2 n \right) \sqrt{\epsilon}} - \frac{i \sqrt{2} \sqrt{\epsilon} l n + m^2 m \frac{\partial}{\partial t} n}{4 \left(\epsilon e^{(\frac{1}{2} M)} l n^2 + e^{(\frac{1}{2} M)} m^2 n \right) \sqrt{\epsilon}} \end{aligned}$$

$$\alpha = -\frac{\sqrt{2}\epsilon\ln^2\frac{\partial}{\partial z}M}{8\left(\epsilon e^{(\frac{1}{2}M)}\ln^2 + e^{(\frac{1}{2}M)}m^2n\right)} + \frac{i\sqrt{2}\sqrt{\epsilon}\ln^2\frac{\partial}{\partial t}M}{8\left(\epsilon e^{(\frac{1}{2}M)}\ln^2 + e^{(\frac{1}{2}M)}m^2n\right)} - \frac{\sqrt{2}m^2n\frac{\partial}{\partial z}M}{8\left(\epsilon e^{(\frac{1}{2}M)}\ln^2 + e^{(\frac{1}{2}M)}m^2n\right)} + \frac{i\sqrt{2}m^2n\frac{\partial}{\partial t}M}{8\left(\epsilon e^{(\frac{1}{2}M)}\ln^2 + e^{(\frac{1}{2}M)}m^2n\right)\sqrt{\epsilon}} \\ - \frac{\sqrt{2}\sqrt{\epsilon}\ln + m^2n\frac{\partial}{\partial z}m}{8\left(\epsilon e^{(\frac{1}{2}M)}\ln^2 + e^{(\frac{1}{2}M)}m^2n\right)} + \frac{\sqrt{2}\sqrt{\epsilon}\ln + m^2m\frac{\partial}{\partial z}n}{8\left(\epsilon e^{(\frac{1}{2}M)}\ln^2 + e^{(\frac{1}{2}M)}m^2n\right)} + \frac{i\sqrt{2}\sqrt{\epsilon}\ln + m^2n\frac{\partial}{\partial t}m}{8\left(\epsilon e^{(\frac{1}{2}M)}\ln^2 + e^{(\frac{1}{2}M)}m^2n\right)\sqrt{\epsilon}} - \frac{i\sqrt{2}\sqrt{\epsilon}\ln + m^2m\frac{\partial}{\partial t}n}{8\left(\epsilon e^{(\frac{1}{2}M)}\ln^2 + e^{(\frac{1}{2}M)}m^2n\right)\sqrt{\epsilon}}$$

$$\begin{aligned} \beta = & \frac{\sqrt{2}\epsilon \ln^2 \frac{\partial}{\partial z} M}{8 \left(\epsilon e^{(\frac{1}{2} M)} \ln^2 + e^{(\frac{1}{2} M)} m^2 n \right)} + \frac{i \sqrt{2} \sqrt{\epsilon} \ln^2 \frac{\partial}{\partial t} M}{8 \left(\epsilon e^{(\frac{1}{2} M)} \ln^2 + e^{(\frac{1}{2} M)} m^2 n \right)} + \frac{\sqrt{2} m^2 n \frac{\partial}{\partial z} M}{8 \left(\epsilon e^{(\frac{1}{2} M)} \ln^2 + e^{(\frac{1}{2} M)} m^2 n \right)} + \frac{i \sqrt{2} m^2 n \frac{\partial}{\partial t} M}{8 \left(\epsilon e^{(\frac{1}{2} M)} \ln^2 + e^{(\frac{1}{2} M)} m^2 n \right)} \sqrt{\epsilon} \\ & - \frac{\sqrt{2} \sqrt{\epsilon} \ln + m^2 n \frac{\partial}{\partial z} m}{8 \left(\epsilon e^{(\frac{1}{2} M)} \ln^2 + e^{(\frac{1}{2} M)} m^2 n \right)} + \frac{\sqrt{2} \sqrt{\epsilon} \ln + m^2 m \frac{\partial}{\partial z} n}{8 \left(\epsilon e^{(\frac{1}{2} M)} \ln^2 + e^{(\frac{1}{2} M)} m^2 n \right)} - \frac{i \sqrt{2} \sqrt{\epsilon} \ln + m^2 n \frac{\partial}{\partial t} m}{8 \left(\epsilon e^{(\frac{1}{2} M)} \ln^2 + e^{(\frac{1}{2} M)} m^2 n \right)} \sqrt{\epsilon} + \frac{i \sqrt{2} \sqrt{\epsilon} \ln + m^2 m \frac{\partial}{\partial t} n}{8 \left(\epsilon e^{(\frac{1}{2} M)} \ln^2 + e^{(\frac{1}{2} M)} m^2 n \right)} \sqrt{\epsilon} \end{aligned}$$

$$\tau = -\frac{\sqrt{2}\epsilon n \frac{\partial}{\partial z} l}{8\left(\epsilon e^{(\frac{1}{2}M)}ln + e^{(\frac{1}{2}M)}m^2\right)} - \frac{\sqrt{2}\epsilon l \frac{\partial}{\partial z} n}{8\left(\epsilon e^{(\frac{1}{2}M)}ln + e^{(\frac{1}{2}M)}m^2\right)} - \frac{i\sqrt{2}\sqrt{\epsilon}n \frac{\partial}{\partial t} l}{8\left(\epsilon e^{(\frac{1}{2}M)}ln + e^{(\frac{1}{2}M)}m^2\right)} \\ - \frac{i\sqrt{2}\sqrt{\epsilon}l \frac{\partial}{\partial t} n}{8\left(\epsilon e^{(\frac{1}{2}M)}ln + e^{(\frac{1}{2}M)}m^2\right)} - \frac{\sqrt{2}m \frac{\partial}{\partial z} m}{4\left(\epsilon e^{(\frac{1}{2}M)}ln + e^{(\frac{1}{2}M)}m^2\right)} - \frac{i\sqrt{2}m \frac{\partial}{\partial t} m}{4\left(\epsilon e^{(\frac{1}{2}M)}ln + e^{(\frac{1}{2}M)}m^2\right)}\sqrt{\epsilon}$$

$$\begin{aligned} \pi = & \frac{\sqrt{2}\epsilon n \frac{\partial}{\partial z} l}{8 \left(\epsilon e^{(\frac{1}{2} M)} l n + e^{(\frac{1}{2} M)} m^2 \right)} + \frac{\sqrt{2}\epsilon l \frac{\partial}{\partial z} n}{8 \left(\epsilon e^{(\frac{1}{2} M)} l n + e^{(\frac{1}{2} M)} m^2 \right)} - \frac{i \sqrt{2}\sqrt{\epsilon} n \frac{\partial}{\partial t} l}{8 \left(\epsilon e^{(\frac{1}{2} M)} l n + e^{(\frac{1}{2} M)} m^2 \right)} \\ & - \frac{i \sqrt{2}\sqrt{\epsilon} l \frac{\partial}{\partial t} n}{8 \left(\epsilon e^{(\frac{1}{2} M)} l n + e^{(\frac{1}{2} M)} m^2 \right)} + \frac{\sqrt{2} m \frac{\partial}{\partial z} m}{4 \left(\epsilon e^{(\frac{1}{2} M)} l n + e^{(\frac{1}{2} M)} m^2 \right)} - \frac{i \sqrt{2} m \frac{\partial}{\partial t} m}{4 \left(\epsilon e^{(\frac{1}{2} M)} l n + e^{(\frac{1}{2} M)} m^2 \right)} \sqrt{\epsilon} \end{aligned}$$

$$\epsilon = 0$$

$$\gamma = 0$$

Ricci Tensor Components

$$\Phi_{10} = 0$$

$$\begin{aligned}\Phi_{11} = & -\frac{\epsilon^2 l n \frac{\partial^2}{(\partial z)^2} M}{8(\epsilon^2 e^M l n + \epsilon e^M m^2)} - \frac{\epsilon l n \frac{\partial^2}{(\partial t)^2} M}{8(\epsilon^2 e^M l n + \epsilon e^M m^2)} - \frac{\epsilon m^2 \frac{\partial^2}{(\partial z)^2} M}{8(\epsilon^2 e^M l n + \epsilon e^M m^2)} + \frac{\epsilon^2 \frac{\partial}{\partial z} l \frac{\partial}{\partial z} n}{16(\epsilon^2 e^M l n + \epsilon e^M m^2)} \\ & - \frac{m^2 \frac{\partial^2}{(\partial t)^2} M}{8(\epsilon^2 e^M l n + \epsilon e^M m^2)} + \frac{\epsilon \frac{\partial}{\partial z} m^2}{16(\epsilon^2 e^M l n + \epsilon e^M m^2)} + \frac{\epsilon \frac{\partial}{\partial t} l \frac{\partial}{\partial t} n}{16(\epsilon^2 e^M l n + \epsilon e^M m^2)} + \frac{\frac{\partial}{\partial t} m^2}{16(\epsilon^2 e^M l n + \epsilon e^M m^2)}\end{aligned}$$

$$\Phi_{12} = 0$$

$$\begin{aligned}\Phi_{20} = & \frac{\epsilon^3 l n^2 \frac{\partial}{\partial z} M \frac{\partial}{\partial z} l}{8(\epsilon^3 e^M l^2 n^2 + 2\epsilon^2 e^M l m^2 n + \epsilon e^M m^4)} + \frac{\epsilon^3 l^2 n \frac{\partial}{\partial z} M \frac{\partial}{\partial z} n}{8(\epsilon^3 e^M l^2 n^2 + 2\epsilon^2 e^M l m^2 n + \epsilon e^M m^4)} - \frac{i \epsilon^{\frac{5}{2}} l n^2 \frac{\partial}{\partial z} M \frac{\partial}{\partial t} l}{8(\epsilon^3 e^M l^2 n^2 + 2\epsilon^2 e^M l m^2 n + \epsilon e^M m^4)} - \frac{i \epsilon^{\frac{5}{2}} l n^2 \frac{\partial}{\partial t} M \frac{\partial}{\partial z} l}{8(\epsilon^3 e^M l^2 n^2 + 2\epsilon^2 e^M l m^2 n + \epsilon e^M m^4)} \\ & - \frac{i \epsilon^{\frac{5}{2}} l^2 n \frac{\partial}{\partial z} M \frac{\partial}{\partial t} n}{8(\epsilon^3 e^M l^2 n^2 + 2\epsilon^2 e^M l m^2 n + \epsilon e^M m^4)} - \frac{i \epsilon^{\frac{5}{2}} l^2 n \frac{\partial}{\partial t} M \frac{\partial}{\partial z} n}{8(\epsilon^3 e^M l^2 n^2 + 2\epsilon^2 e^M l m^2 n + \epsilon e^M m^4)} - \frac{\epsilon^2 l n^2 \frac{\partial}{\partial t} M \frac{\partial}{\partial t} l}{8(\epsilon^3 e^M l^2 n^2 + 2\epsilon^2 e^M l m^2 n + \epsilon e^M m^4)} + \frac{\epsilon^2 m^2 n \frac{\partial}{\partial z} M \frac{\partial}{\partial z} l}{8(\epsilon^3 e^M l^2 n^2 + 2\epsilon^2 e^M l m^2 n + \epsilon e^M m^4)} \\ & + \frac{\epsilon^3 n^2 \frac{\partial}{\partial z} l^2}{16(\epsilon^3 e^M l^2 n^2 + 2\epsilon^2 e^M l m^2 n + \epsilon e^M m^4)} - \frac{\epsilon^3 l n^2 \frac{\partial^2}{(\partial z)^2} l}{8(\epsilon^3 e^M l^2 n^2 + 2\epsilon^2 e^M l m^2 n + \epsilon e^M m^4)} + \frac{\epsilon^2 l m n \frac{\partial}{\partial z} M \frac{\partial}{\partial z} m}{4(\epsilon^3 e^M l^2 n^2 + 2\epsilon^2 e^M l m^2 n + \epsilon e^M m^4)} - \frac{\epsilon^2 l^2 n \frac{\partial}{\partial t} M \frac{\partial}{\partial t} n}{8(\epsilon^3 e^M l^2 n^2 + 2\epsilon^2 e^M l m^2 n + \epsilon e^M m^4)} \\ & + \frac{\epsilon^2 l m^2 \frac{\partial}{\partial z} M \frac{\partial}{\partial z} n}{8(\epsilon^3 e^M l^2 n^2 + 2\epsilon^2 e^M l m^2 n + \epsilon e^M m^4)} + \frac{\epsilon^3 l^2 \frac{\partial}{\partial z} n^2}{16(\epsilon^3 e^M l^2 n^2 + 2\epsilon^2 e^M l m^2 n + \epsilon e^M m^4)} - \frac{\epsilon^3 l^2 n \frac{\partial^2}{(\partial z)^2} n}{8(\epsilon^3 e^M l^2 n^2 + 2\epsilon^2 e^M l m^2 n + \epsilon e^M m^4)} - \frac{i \epsilon^{\frac{3}{2}} m^2 n \frac{\partial}{\partial z} M \frac{\partial}{\partial t} l}{8(\epsilon^3 e^M l^2 n^2 + 2\epsilon^2 e^M l m^2 n + \epsilon e^M m^4)} \\ & + \frac{i \epsilon^{\frac{5}{2}} l n^2 \frac{\partial^2}{\partial t \partial z} l}{4(\epsilon^3 e^M l^2 n^2 + 2\epsilon^2 e^M l m^2 n + \epsilon e^M m^4)} - \frac{i \epsilon^{\frac{3}{2}} m^2 n \frac{\partial}{\partial t} M \frac{\partial}{\partial z} l}{8(\epsilon^3 e^M l^2 n^2 + 2\epsilon^2 e^M l m^2 n + \epsilon e^M m^4)} - \frac{i \epsilon^{\frac{5}{2}} n^2 \frac{\partial}{\partial t} l \frac{\partial}{\partial z} l}{8(\epsilon^3 e^M l^2 n^2 + 2\epsilon^2 e^M l m^2 n + \epsilon e^M m^4)} - \frac{i \epsilon^{\frac{3}{2}} l m n \frac{\partial}{\partial z} M \frac{\partial}{\partial t} m}{4(\epsilon^3 e^M l^2 n^2 + 2\epsilon^2 e^M l m^2 n + \epsilon e^M m^4)} \\ & - \frac{i \epsilon^{\frac{3}{2}} l m n \frac{\partial}{\partial t} M \frac{\partial}{\partial z} m}{4(\epsilon^3 e^M l^2 n^2 + 2\epsilon^2 e^M l m^2 n + \epsilon e^M m^4)} - \frac{i \epsilon^{\frac{3}{2}} l m^2 \frac{\partial}{\partial z} M \frac{\partial}{\partial t} n}{8(\epsilon^3 e^M l^2 n^2 + 2\epsilon^2 e^M l m^2 n + \epsilon e^M m^4)} + \frac{i \epsilon^{\frac{5}{2}} l^2 n \frac{\partial^2}{\partial t \partial z} n}{4(\epsilon^3 e^M l^2 n^2 + 2\epsilon^2 e^M l m^2 n + \epsilon e^M m^4)} - \frac{i \epsilon^{\frac{3}{2}} l m^2 \frac{\partial}{\partial t} M \frac{\partial}{\partial z} n}{8(\epsilon^3 e^M l^2 n^2 + 2\epsilon^2 e^M l m^2 n + \epsilon e^M m^4)} - \dots\end{aligned}$$

$$\Phi_{21} = 0$$

$$\begin{aligned}
\Phi_{22} = & \frac{\epsilon^3 n^3 \frac{\partial}{\partial z} l^2}{16(\epsilon^3 e^M l^2 n^3 + 2\epsilon^2 e^M l m^2 n^2 + \epsilon e^M m^4 n)} - \frac{\epsilon^3 l n^3 \frac{\partial^2}{(\partial z)^2} l}{8(\epsilon^3 e^M l^2 n^3 + 2\epsilon^2 e^M l m^2 n^2 + \epsilon e^M m^4 n)} - \frac{\epsilon^3 l^2 n \frac{\partial}{\partial z} n^2}{16(\epsilon^3 e^M l^2 n^3 + 2\epsilon^2 e^M l m^2 n^2 + \epsilon e^M m^4 n)} \\
& + \frac{\epsilon^3 l^2 n^2 \frac{\partial^2}{(\partial z)^2} n}{8(\epsilon^3 e^M l^2 n^3 + 2\epsilon^2 e^M l m^2 n^2 + \epsilon e^M m^4 n)} + \frac{\epsilon^2 n^3 \frac{\partial}{\partial t} l^2}{16(\epsilon^3 e^M l^2 n^3 + 2\epsilon^2 e^M l m^2 n^2 + \epsilon e^M m^4 n)} \\
& - \frac{\epsilon^2 l n^3 \frac{\partial^2}{(\partial t)^2} l}{8(\epsilon^3 e^M l^2 n^3 + 2\epsilon^2 e^M l m^2 n^2 + \epsilon e^M m^4 n)} - \frac{\epsilon^2 m^2 n^2 \frac{\partial^2}{(\partial z)^2} l}{8(\epsilon^3 e^M l^2 n^3 + 2\epsilon^2 e^M l m^2 n^2 + \epsilon e^M m^4 n)} + \frac{\epsilon^2 m n^2 \frac{\partial}{\partial z} l \frac{\partial}{\partial z} m}{4(\epsilon^3 e^M l^2 n^3 + 2\epsilon^2 e^M l m^2 n^2 + \epsilon e^M m^4 n)} \\
& - \frac{\epsilon^2 l m n^2 \frac{\partial^2}{(\partial z)^2} m}{4(\epsilon^3 e^M l^2 n^3 + 2\epsilon^2 e^M l m^2 n^2 + \epsilon e^M m^4 n)} - \frac{\epsilon^2 l^2 n \frac{\partial}{\partial t} n^2}{16(\epsilon^3 e^M l^2 n^3 + 2\epsilon^2 e^M l m^2 n^2 + \epsilon e^M m^4 n)} + \frac{\epsilon^2 l^2 n^2 \frac{\partial^2}{(\partial t)^2} n}{8(\epsilon^3 e^M l^2 n^3 + 2\epsilon^2 e^M l m^2 n^2 + \epsilon e^M m^4 n)} \\
& - \frac{\epsilon^2 m^2 n \frac{\partial}{\partial z} l \frac{\partial}{\partial z} n}{8(\epsilon^3 e^M l^2 n^3 + 2\epsilon^2 e^M l m^2 n^2 + \epsilon e^M m^4 n)} - \frac{\epsilon^2 l m^2 \frac{\partial}{\partial z} n^2}{8(\epsilon^3 e^M l^2 n^3 + 2\epsilon^2 e^M l m^2 n^2 + \epsilon e^M m^4 n)} + \frac{3\epsilon^2 l m^2 n \frac{\partial^2}{(\partial z)^2} n}{8(\epsilon^3 e^M l^2 n^3 + 2\epsilon^2 e^M l m^2 n^2 + \epsilon e^M m^4 n)} \\
& + \frac{\sqrt{\epsilon l n + m^2} \epsilon^2 n^2 \frac{\partial}{\partial z} l \frac{\partial}{\partial z} m}{8(\epsilon^3 e^M l^2 n^3 + 2\epsilon^2 e^M l m^2 n^2 + \epsilon e^M m^4 n)} - \frac{\sqrt{\epsilon l n + m^2} \epsilon^2 l n^2 \frac{\partial^2}{(\partial z)^2} m}{4(\epsilon^3 e^M l^2 n^3 + 2\epsilon^2 e^M l m^2 n^2 + \epsilon e^M m^4 n)} - \frac{\sqrt{\epsilon l n + m^2} \epsilon^2 m n \frac{\partial}{\partial z} l \frac{\partial}{\partial z} n}{8(\epsilon^3 e^M l^2 n^3 + 2\epsilon^2 e^M l m^2 n^2 + \epsilon e^M m^4 n)} \\
& + \frac{\sqrt{\epsilon l n + m^2} \epsilon^2 l n \frac{\partial}{\partial z} m \frac{\partial}{\partial z} n}{8(\epsilon^3 e^M l^2 n^3 + 2\epsilon^2 e^M l m^2 n^2 + \epsilon e^M m^4 n)} - \frac{\sqrt{\epsilon l n + m^2} \epsilon^2 l m \frac{\partial}{\partial z} n^2}{8(\epsilon^3 e^M l^2 n^3 + 2\epsilon^2 e^M l m^2 n^2 + \epsilon e^M m^4 n)} + \frac{\sqrt{\epsilon l n + m^2} \epsilon^2 l m n \frac{\partial^2}{(\partial z)^2} n}{4(\epsilon^3 e^M l^2 n^3 + 2\epsilon^2 e^M l m^2 n^2 + \epsilon e^M m^4 n)} \\
& - \frac{\epsilon m^2 n^2 \frac{\partial^2}{(\partial t)^2} l}{8(\epsilon^3 e^M l^2 n^3 + 2\epsilon^2 e^M l m^2 n^2 + \epsilon e^M m^4 n)} + \frac{\epsilon m n^2 \frac{\partial}{\partial t} l \frac{\partial}{\partial t} m}{4(\epsilon^3 e^M l^2 n^3 + 2\epsilon^2 e^M l m^2 n^2 + \epsilon e^M m^4 n)} - \frac{\epsilon l m n^2 \frac{\partial^2}{(\partial t)^2} m}{4(\epsilon^3 e^M l^2 n^3 + 2\epsilon^2 e^M l m^2 n^2 + \epsilon e^M m^4 n)} \\
& + \frac{\epsilon m^2 n \frac{\partial}{\partial z} m^2}{4(\epsilon^3 e^M l^2 n^3 + 2\epsilon^2 e^M l m^2 n^2 + \epsilon e^M m^4 n)} - \frac{\epsilon m^3 n \frac{\partial^2}{(\partial z)^2} m}{4(\epsilon^3 e^M l^2 n^3 + 2\epsilon^2 e^M l m^2 n^2 + \epsilon e^M m^4 n)} - \dots
\end{aligned}$$

$$\Psi_1 = 0$$

$$\begin{aligned} \Psi_2 = & \frac{\epsilon^3 l^2 n^2 \frac{\partial^2}{(\partial z)^2} M}{12(\epsilon^3 e^M l^2 n^2 + 2\epsilon^2 e^M l m^2 n + \epsilon e^M m^4)} + \frac{\epsilon^2 l^2 n^2 \frac{\partial^2}{(\partial t)^2} M}{12(\epsilon^3 e^M l^2 n^2 + 2\epsilon^2 e^M l m^2 n + \epsilon e^M m^4)} + \frac{\epsilon^2 l m^2 n \frac{\partial^2}{(\partial z)^2} M}{6(\epsilon^3 e^M l^2 n^2 + 2\epsilon^2 e^M l m^2 n + \epsilon e^M m^4)} + \frac{\epsilon^3 n^2 \frac{\partial}{\partial z} l^2}{48(\epsilon^3 e^M l^2 n^2 + 2\epsilon^2 e^M l m^2 n + \epsilon e^M m^4)} \\ & - \frac{\epsilon^3 l n^2 \frac{\partial^2}{(\partial z)^2} l}{24(\epsilon^3 e^M l^2 n^2 + 2\epsilon^2 e^M l m^2 n + \epsilon e^M m^4)} + \frac{\epsilon^3 l n \frac{\partial}{\partial z} l \frac{\partial}{\partial z} n}{24(\epsilon^3 e^M l^2 n^2 + 2\epsilon^2 e^M l m^2 n + \epsilon e^M m^4)} + \frac{\epsilon^3 l^2 \frac{\partial}{\partial z} n^2}{48(\epsilon^3 e^M l^2 n^2 + 2\epsilon^2 e^M l m^2 n + \epsilon e^M m^4)} - \frac{\epsilon^3 l^2 n \frac{\partial^2}{(\partial z)^2} n}{24(\epsilon^3 e^M l^2 n^2 + 2\epsilon^2 e^M l m^2 n + \epsilon e^M m^4)} \\ & + \frac{\epsilon l m^2 n \frac{\partial^2}{(\partial t)^2} M}{6(\epsilon^3 e^M l^2 n^2 + 2\epsilon^2 e^M l m^2 n + \epsilon e^M m^4)} + \frac{\epsilon m^4 \frac{\partial^2}{(\partial z)^2} M}{12(\epsilon^3 e^M l^2 n^2 + 2\epsilon^2 e^M l m^2 n + \epsilon e^M m^4)} + \frac{\epsilon^2 n^2 \frac{\partial}{\partial t} l^2}{48(\epsilon^3 e^M l^2 n^2 + 2\epsilon^2 e^M l m^2 n + \epsilon e^M m^4)} - \frac{\epsilon^2 l n^2 \frac{\partial^2}{(\partial t)^2} l}{24(\epsilon^3 e^M l^2 n^2 + 2\epsilon^2 e^M l m^2 n + \epsilon e^M m^4)} \\ & - \frac{\epsilon^2 m^2 n \frac{\partial^2}{(\partial z)^2} l}{24(\epsilon^3 e^M l^2 n^2 + 2\epsilon^2 e^M l m^2 n + \epsilon e^M m^4)} + \frac{\epsilon^2 m n \frac{\partial}{\partial z} l \frac{\partial}{\partial z} m}{12(\epsilon^3 e^M l^2 n^2 + 2\epsilon^2 e^M l m^2 n + \epsilon e^M m^4)} - \frac{\epsilon^2 l m n \frac{\partial^2}{(\partial z)^2} m}{12(\epsilon^3 e^M l^2 n^2 + 2\epsilon^2 e^M l m^2 n + \epsilon e^M m^4)} + \frac{\epsilon^2 l n \frac{\partial}{\partial t} l \frac{\partial}{\partial t} n}{24(\epsilon^3 e^M l^2 n^2 + 2\epsilon^2 e^M l m^2 n + \epsilon e^M m^4)} \\ & + \frac{\epsilon^2 l^2 \frac{\partial}{\partial t} n^2}{48(\epsilon^3 e^M l^2 n^2 + 2\epsilon^2 e^M l m^2 n + \epsilon e^M m^4)} - \frac{\epsilon^2 l^2 n \frac{\partial^2}{(\partial t)^2} n}{24(\epsilon^3 e^M l^2 n^2 + 2\epsilon^2 e^M l m^2 n + \epsilon e^M m^4)} + \frac{\epsilon^2 l m \frac{\partial}{\partial z} m \frac{\partial}{\partial z} n}{12(\epsilon^3 e^M l^2 n^2 + 2\epsilon^2 e^M l m^2 n + \epsilon e^M m^4)} - \frac{\epsilon^2 l m^2 \frac{\partial^2}{(\partial z)^2} n}{24(\epsilon^3 e^M l^2 n^2 + 2\epsilon^2 e^M l m^2 n + \epsilon e^M m^4)} \\ & + \frac{m^4 \frac{\partial^2}{(\partial t)^2} M}{12(\epsilon^3 e^M l^2 n^2 + 2\epsilon^2 e^M l m^2 n + \epsilon e^M m^4)} - \frac{\epsilon m^2 n \frac{\partial^2}{(\partial t)^2} l}{24(\epsilon^3 e^M l^2 n^2 + 2\epsilon^2 e^M l m^2 n + \epsilon e^M m^4)} + \frac{\epsilon m n \frac{\partial}{\partial t} l \frac{\partial}{\partial t} m}{12(\epsilon^3 e^M l^2 n^2 + 2\epsilon^2 e^M l m^2 n + \epsilon e^M m^4)} + \frac{i \sqrt{\epsilon l n} + m^2 \epsilon^{\frac{3}{2}} n \frac{\partial}{\partial z} l \frac{\partial}{\partial t} m}{8(\epsilon^3 e^M l^2 n^2 + 2\epsilon^2 e^M l m^2 n + \epsilon e^M m^4)} \\ & - \frac{\epsilon l m n \frac{\partial^2}{(\partial t)^2} m}{12(\epsilon^3 e^M l^2 n^2 + 2\epsilon^2 e^M l m^2 n + \epsilon e^M m^4)} - \frac{i \sqrt{\epsilon l n} + m^2 \epsilon^{\frac{3}{2}} n \frac{\partial}{\partial t} l \frac{\partial}{\partial z} m}{8(\epsilon^3 e^M l^2 n^2 + 2\epsilon^2 e^M l m^2 n + \epsilon e^M m^4)} + \frac{\epsilon m^2 \frac{\partial}{\partial z} m^2}{12(\epsilon^3 e^M l^2 n^2 + 2\epsilon^2 e^M l m^2 n + \epsilon e^M m^4)} - \dots \end{aligned}$$

$$\Psi_3 = 0$$

$$\begin{aligned}
\Psi_4 = & \frac{\epsilon^3 l n^3 \frac{\partial}{\partial z} M \frac{\partial}{\partial z} l}{8(\epsilon^3 e^M l^2 n^3 + 2\epsilon^2 e^M l m^2 n^2 + \epsilon e^M m^4 n)} - \frac{\epsilon^3 l^2 n^2 \frac{\partial}{\partial z} M \frac{\partial}{\partial z} n}{8(\epsilon^3 e^M l^2 n^3 + 2\epsilon^2 e^M l m^2 n^2 + \epsilon e^M m^4 n)} - \frac{i \epsilon^{\frac{5}{2}} l n^3 \frac{\partial}{\partial z} M \frac{\partial}{\partial t} l}{8(\epsilon^3 e^M l^2 n^3 + 2\epsilon^2 e^M l m^2 n^2 + \epsilon e^M m^4 n)} \\
& - \frac{i \epsilon^{\frac{5}{2}} l n^3 \frac{\partial}{\partial t} M \frac{\partial}{\partial z} l}{8(\epsilon^3 e^M l^2 n^3 + 2\epsilon^2 e^M l m^2 n^2 + \epsilon e^M m^4 n)} + \frac{i \epsilon^{\frac{5}{2}} l^2 n^2 \frac{\partial}{\partial z} M \frac{\partial}{\partial t} n}{8(\epsilon^3 e^M l^2 n^3 + 2\epsilon^2 e^M l m^2 n^2 + \epsilon e^M m^4 n)} + \frac{i \epsilon^{\frac{5}{2}} l^2 n^2 \frac{\partial}{\partial t} M \frac{\partial}{\partial z} n}{8(\epsilon^3 e^M l^2 n^3 + 2\epsilon^2 e^M l m^2 n^2 + \epsilon e^M m^4 n)} \\
& - \frac{\epsilon^2 l n^3 \frac{\partial}{\partial t} M \frac{\partial}{\partial t} l}{8(\epsilon^3 e^M l^2 n^3 + 2\epsilon^2 e^M l m^2 n^2 + \epsilon e^M m^4 n)} + \frac{\epsilon^2 m^2 n^2 \frac{\partial}{\partial z} M \frac{\partial}{\partial z} l}{8(\epsilon^3 e^M l^2 n^3 + 2\epsilon^2 e^M l m^2 n^2 + \epsilon e^M m^4 n)} + \frac{\epsilon^3 n^3 \frac{\partial}{\partial z} l^2}{16(\epsilon^3 e^M l^2 n^3 + 2\epsilon^2 e^M l m^2 n^2 + \epsilon e^M m^4 n)} \\
& - \frac{\epsilon^3 l n^3 \frac{\partial^2}{(\partial z)^2} l}{8(\epsilon^3 e^M l^2 n^3 + 2\epsilon^2 e^M l m^2 n^2 + \epsilon e^M m^4 n)} + \frac{\epsilon^2 l m n^2 \frac{\partial}{\partial z} M \frac{\partial}{\partial z} m}{4(\epsilon^3 e^M l^2 n^3 + 2\epsilon^2 e^M l m^2 n^2 + \epsilon e^M m^4 n)} + \frac{\epsilon^2 l^2 n^2 \frac{\partial}{\partial t} M \frac{\partial}{\partial t} n}{8(\epsilon^3 e^M l^2 n^3 + 2\epsilon^2 e^M l m^2 n^2 + \epsilon e^M m^4 n)} \\
& - \frac{3 \epsilon^2 l m^2 n \frac{\partial}{\partial z} M \frac{\partial}{\partial z} n}{8(\epsilon^3 e^M l^2 n^3 + 2\epsilon^2 e^M l m^2 n^2 + \epsilon e^M m^4 n)} - \frac{\epsilon^3 l^2 n \frac{\partial}{\partial z} n^2}{16(\epsilon^3 e^M l^2 n^3 + 2\epsilon^2 e^M l m^2 n^2 + \epsilon e^M m^4 n)} + \frac{\epsilon^3 l^2 n^2 \frac{\partial^2}{(\partial z)^2} n}{8(\epsilon^3 e^M l^2 n^3 + 2\epsilon^2 e^M l m^2 n^2 + \epsilon e^M m^4 n)} \\
& - \frac{i \epsilon^{\frac{3}{2}} m^2 n^2 \frac{\partial}{\partial z} M \frac{\partial}{\partial t} l}{8(\epsilon^3 e^M l^2 n^3 + 2\epsilon^2 e^M l m^2 n^2 + \epsilon e^M m^4 n)} + \frac{i \epsilon^{\frac{5}{2}} l n^3 \frac{\partial^2}{\partial t \partial z} l}{4(\epsilon^3 e^M l^2 n^3 + 2\epsilon^2 e^M l m^2 n^2 + \epsilon e^M m^4 n)} - \frac{i \epsilon^{\frac{3}{2}} m^2 n^2 \frac{\partial}{\partial t} M \frac{\partial}{\partial z} l}{8(\epsilon^3 e^M l^2 n^3 + 2\epsilon^2 e^M l m^2 n^2 + \epsilon e^M m^4 n)} \\
& - \frac{i \epsilon^{\frac{5}{2}} n^3 \frac{\partial}{\partial t} l \frac{\partial}{\partial z} l}{8(\epsilon^3 e^M l^2 n^3 + 2\epsilon^2 e^M l m^2 n^2 + \epsilon e^M m^4 n)} - \frac{i \epsilon^{\frac{3}{2}} l m n^2 \frac{\partial}{\partial z} M \frac{\partial}{\partial t} m}{4(\epsilon^3 e^M l^2 n^3 + 2\epsilon^2 e^M l m^2 n^2 + \epsilon e^M m^4 n)} - \frac{i \epsilon^{\frac{3}{2}} l m n^2 \frac{\partial}{\partial t} M \frac{\partial}{\partial z} m}{4(\epsilon^3 e^M l^2 n^3 + 2\epsilon^2 e^M l m^2 n^2 + \epsilon e^M m^4 n)} \\
& + \frac{\sqrt{\epsilon} l n + m^2 \epsilon^2 l n^2 \frac{\partial}{\partial z} M \frac{\partial}{\partial z} m}{4(\epsilon^3 e^M l^2 n^3 + 2\epsilon^2 e^M l m^2 n^2 + \epsilon e^M m^4 n)} + \frac{3 i \epsilon^{\frac{3}{2}} l m^2 n \frac{\partial}{\partial z} M \frac{\partial}{\partial t} n}{8(\epsilon^3 e^M l^2 n^3 + 2\epsilon^2 e^M l m^2 n^2 + \epsilon e^M m^4 n)} - \dots
\end{aligned}$$

Newman-Penrose Equations

Equations are vanished.

Bianchi Identities

Equations are vanished.

Petrov Type

Type "I"